AI-Powered Civil Law Assistance Platform for Sri Lankan Citizens

25-26J-553

Project Proposal Report

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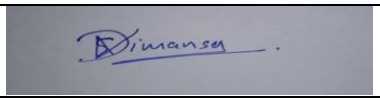
Declaration

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

	Signature	**Date**
Vimansa D. A. S		29-09-2025
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Abstract

LegalEase LK is a legal assistance platform powered by AI, aimed at enhancing access to legal information for everyday citizens in Sri Lanka. Current platforms like LankaLAW and LawMate mainly cater to legal practitioners and lack straightforward, multilingual legal clarifications appropriate for the average person. This study seeks to address this issue by creating an accessible platform that delivers straightforward legal advice in Sinhala and English via a chatbot powered by a large language model (LLM), as well as a lawyer referral system that assists users in locating suitable legal experts based on specialization and geography. The backend utilizes Node.js/Express, while the frontend employs React, guaranteeing scalability, performance, and user-friendliness. The system relies on selected legal data sources like "Edineda Jiwithayata Neethiya" to produce precise and comprehensible answers. The anticipated result is heightening legal awareness, better accessibility to lawyers, and strengthening public confidence in the justice system.

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1. Introduction

Legal information remains inaccessible for the majority of Sri Lankan citizens. Many have to resort to seeking help from professionals because of complicated terminology and lack of knowledge of court processes. The public finds it difficult to understand their rights, what procedure to follow in any given situation, or when professional legal assistance is required. Examples of online legal facilities, such as LankaLAW and LawMate, exist in Sri Lanka. However, all such current solutions are targeted at lawyers, researchers, and legal professionals. Both sites facilitate access to legal databases, case-law research, and professional tools, without offering any simplified explanations or multilingual support suitable for the public. Thus, the majority of the population still has no way of accessing understandable and accessible information on legal issues.

The development of LegalEase LK requires insight into multiple areas of technology and research. First, there is the requirement for Artificial Intelligence Language Models to allow the system to respond naturally to the user and summarize complex legal concepts in a layman's language. Information Retrieval techniques need to be applied to obtain accurate content from Sri Lankan legal sources, and this requires Multilingual NLP to support Sinhala and English queries effectively. Modern web technologies like Node.js and Express have been used for backend development, while React.js has been used to create a user-friendly frontend. Recommendation system techniques will also be necessary in order to implement a lawyer-recommending mechanism that recommends lawyers based on their specialization and location.

Current advanced legal systems in Sri Lanka show strong technological development within the professional legal area. LankaLAW is an AI-driven legal research platform for professionals, while LawMate offers automated legal insights targeted at lawyers and researchers. Although these are some of the state-of-the-art digital legal solution platforms, they do not service the everyday citizen. They lack explanations in simple Sinhala or Tamil and offer no conversational interfaces to help users understand the law in plain language. Furthermore, none of them have a lawyer discovery or recommendation feature devised to assist the public in reaching the right legal assistance.

These limitations indicate flaws in current solutions. The legal vocabulary used in professional sites is incomprehensible by most ordinary citizens, and thus, citizens cannot learn about their rights independently. Current systems cater to the needs of professionals rather than the public, making a barrier to knowledge for non-legal individuals. Secondly, there does not exist a multilingual AI chatbot in Sri Lanka that could explain the law in a simplified manner suitable for citizens. Neither is there an integrated system that recommends lawyers to the users based on their needs, location, or case type. LegalEase LK overcomes these deficiencies with its all-new creation of a citizen-centric, legal support network. The multilingual LLM-based chatbot is able to explain legal terminologies and jargon in simple language, making such legal knowledge available for all. Initially designed to target both Sinhala and English speakers, with a possible extension into Tamil, the system incorporates a lawyer directory and smart recommendation engine that connects users with suitable lawyers based on specializations and districts. Unlike existing systems, LegalEase LK is designed for ordinary citizens rather than

lawyers, thus offering simplicity, clarity, and usability, against which identified gaps in current solutions are directly matched.

2. Background & Literature survey

Access to legal information in Sri Lanka has long been limited for ordinary citizens, who often struggle to interpret legal documents, understand their fundamental rights, or determine the correct legal actions to take. The complexity of legal language, combined with the lack of simplified explanations, makes legal knowledge difficult to access without professional help. Although several digital legal platforms exist in Sri Lanka, the majority of them—such as LankaLAW and LawMate—are primarily designed for lawyers, researchers, and law students. These platforms offer comprehensive legal databases, case law repositories, and tools for professional legal research. However, they do not provide multilingual support, plain-language explanations, or user-friendly features aimed at non-professional audiences. As a result, the general public remains underserved, and many individuals still rely on indirect, inaccurate, or incomplete legal advice due to the absence of accessible legal guidance tools.
The development of a platform like LegalEase LK therefore requires a strong understanding of existing technological and academic foundations. Artificial Intelligence Language Models (LLMs) play a key role in enabling systems to interpret user queries and generate meaningful legal explanations in conversational form. Modern LLMs have demonstrated significant success in simplifying complex information, summarizing documents, and engaging in dialogue that feels natural to the user. Alongside LLMs, information retrieval techniques are essential for extracting relevant content from Sri Lankan legal sources and structuring it in a way that the AI can use effectively. Multilingual Natural Language Processing is also an important foundation, as the system must operate in Sinhala and English to be truly inclusive for Sri Lankan citizens.

Internationally, AI has been widely adopted in legal domains, leading to the development of platforms such as DoNotPay, LawDroid, and LegalZoom. These services assist with document generation, automated legal responses, and preliminary legal guidance. However, these systems are built around legal frameworks in Western countries and do not apply to Sri Lankan law or local languages. Moreover, their functionality is primarily limited to document automation rather than educational, conversational support that empowers citizens to understand their rights.

Within Sri Lanka, platforms like LankaLAW represent the state of the art in legal research technology. LankaLAW uses AI to assist lawyers in finding case references, statutes, and judgments more efficiently. Similarly, LawMate offers automated legal insights, enabling legal professionals to speed up their research and analysis processes. While these systems incorporate modern technology, they are not designed for the public. They lack simplified explanations, accessible language, and interactive guidance that would help citizens understand legal concepts independently. Importantly, they also do not offer features such as lawyer recommendations or localized guidance that help users identify suitable legal professionals based on location or specialization.

From existing literature and related systems, several research gaps have become evident. There is no Sri Lankan platform that provides simplified, citizen-friendly legal explanations in Sinhala or English through an AI-powered conversational interface. There is no system that combines legal guidance with a lawyer's recommendation mechanism tailored to the user's district and legal needs. Additionally, current tools focus on professionals rather than the public, leaving a wide gap in public legal awareness and accessibility. This creates a clear need for a solution like LegalEase LK, which integrates natural-language legal assistance with multilingual support and smart lawyer recommendations.

LegalEase LK builds on advancements in AI and web technologies to address these gaps. By leveraging LLM-based conversational models, information retrieval systems, and modern web frameworks like React and Node.js, the platform aims to provide ordinary Sri Lankan citizens with direct, simple, and reliable access to legal knowledge. The literature and technology landscape clearly show the absence of such a citizen-centered solution, validating the significance and necessity of this research.

3. Research Gap

Although Sri Lanka has made progress in digitalizing legal information, the existing technological solutions primarily cater to legal professionals rather than ordinary citizens. Platforms such as LankaLAW and LawMate provide access to case law, statutes, and legal analysis tools, but they are designed using professional terminology and assume a level of legal literacy that the general public does not possess. As a result, there is a significant disconnect between available digital legal tools and the everyday user who seeks simple explanations and practical guidance.

Another major gap in the current landscape is the absence of an AI-powered system that can explain legal information in plain, easy-to-understand Sinhala or English. Existing platforms do not offer conversational interfaces or multilingual chatbots capable of breaking down complex legal concepts into language that ordinary citizens can easily grasp. This lack of simplified legal communication contributes to low legal awareness and leaves many individuals dependent on unofficial sources or unverified advice.

Furthermore, no available platform integrates a lawyer recommendation mechanism that matches users with suitable legal professionals based on location and specialization. Citizens, especially in rural areas, often struggle to find the right lawyer and must rely on word of mouth, which can lead to delays, misinformation, or mismatched legal representation. The absence of a digital tool that bridges citizens with the legal community further broadens the accessibility gap.

International legal AI platforms also fail to address the local context. Systems like DoNotPay or LegalZoom focus on Western legal frameworks and provide automated solutions irrelevant to Sri Lankan law, language, or cultural context. Meanwhile, domestic platforms focus heavily on legal research rather than citizen education, leaving no solution tailored specifically to the public's everyday legal questions.

This creates a clear research gap: Sri Lanka lacks a user-centered, multilingual, AI-driven legal assistance platform designed to help ordinary citizens understand their rights and connect with appropriate legal professionals. LegalEase LK aims to fill this gap by offering plain-language explanations, Sinhala–English conversational support, and a smart lawyer recommendation system—features that are currently absent in both local and international alternatives.

4. Research Problem

Ordinary Sri Lankan citizens face significant challenges in accessing and understanding legal information. The majority of legal resources available today are written in complex legal terminology that is difficult for non-professionals to interpret. Although digital legal platforms exist, they are primarily designed for lawyers, researchers, and legal students, offering technical tools and databases rather than simple explanations for the public. These platforms do not provide multilingual, plain-language guidance, nor do they support conversational assistance that can help users navigate legal issues independently.

As a result, citizens often lack awareness of their legal rights and responsibilities, leading to misinformation, delays in seeking help, and an increased reliance on unofficial or unreliable advice. Furthermore, many individuals struggle to identify suitable legal professionals due to the absence of an accessible lawyer recommendation mechanism. This gap is especially prominent in rural areas, where legal resources and guidance are limited.

The core research problem, therefore, is the lack of an accessible, multilingual, AI-powered legal assistance platform that can explain legal concepts in simple language and connect citizens with appropriate lawyers. Without such a system, the general public continues to face barriers in understanding the law, making informed decisions, and accessing justice efficiently.

5. Objectives

Research objectives are concrete statements that describe what the project aims to achieve. These objectives will be evaluated during the final presentation and viva to determine their successful achievement. Each objective follows the **SMART** framework Specific, Measurable, Achievable, Realistic, and Time-bound.

a. Main Objectives

To design and develop an AI-powered, multilingual legal assistance platform that provides Sri Lankan citizens with simple, accessible explanations of legal information and helps them connect with suitable legal professionals.

b. Specific Objectives

- To develop a multilingual chatbot using LLM technology that can explain legal concepts in simple Sinhala and English.
- To build a legal information retrieval mechanism that delivers accurate, relevant, and easy-to-understand responses based on Sri Lankan legal sources.
- To create a lawyer recommendation system that suggests suitable lawyers based on specialization and geographic location.
- To design a user-friendly web interface using React and Node.js that allows citizens to interact easily with the platform and access legal assistance.
- To integrate "Edineda Jiwithayata Neethiya" and other legal datasets into a structured knowledge base suitable for AI processing.
- To improve public legal awareness by providing simplified legal guidance, reducing dependency on unofficial or inaccurate sources.
- To evaluate the system's effectiveness through user testing, focusing on accuracy, usability, and impact on legal understanding.

c. The objectives are SMART

Figure 1.1

6. Methodology

The methodology outlines how the LegalEase LK platform will be designed, developed, and evaluated. It explains the system structure, workflow, data handling approach, and the project timeline required to complete the research objectives.

6.1. System Overview and Workflow

The **system diagram** illustrates the overall flow between users, the web platform, and the AI model.

1. Users enter their legal questions through the web interface.
2. The question is securely sent to the backend and processed by the custom LLM.
3. The AI model converts the question into a vector and retrieves the most relevant legal sections from the vector database.
4. The model summarizes the retrieved law and identifies the legal category related to the user's issue.
5. The Web Platform then uses this category to fetch suitable lawyers from the database.
6. Finally, the platform displays a clear legal explanation along with the best-matched lawyers to the user.
7. Both users and lawyers interact through the same platform, ensuring transparent and AI-supported legal guidance

Main Task	Sub-Tasks
1. Requirement Gathering	Identify user needs, analyze existing systems, and define system goals.
2. Dataset Preparation	Extract content from "Edineda Jiwithayata Neethiya", clean and structure legal text.
3. System Design	Create architecture diagram, workflow diagram, plan APIs and modules.
4. Backend Development	Develop authentication, chatbot API, lawyer recommendation API.
5. Frontend Development	Build UI with React, integrate chatbot UI, lawyer directory
6. AI Chatbot Integration	Implement LLM queries, prompt engineering, and response formatting.
Lawyer Recommendation Module	Categorize lawyers, set filters (district, specialization), implement search logic.
Testing & Evaluation	Conduct functional testing, performance testing, user acceptance testing
Deployment	Host backend and frontend on cloud platforms
Documentation	Prepare proposal, final report, technical documentation

6.2. Data Collection

To build the LegalEase LK knowledge base and lawyer directory, several data sources will be used:

1. **Primary Legal Dataset**
    a. *Edineda Jiwithayata Neethiya* (Part 1, 2, 3)
    b. Provides simplified legal explanations suitable for citizens.
2. **Legal Information Sources**
    a. Publicly available Sri Lankan laws
    b. Edineda Jiwithayata Neethiya (Part 1, 2, 3)
3. **Lawyer Directory Data**
    a. Publicly available lawyer lists
    b. Bar Association directories
4. **User Interaction Data**
    a. Anonymous user queries (for future improvements)
    b. Feedback collected during testing and UAT

All textual data will be cleaned, structured, tokenized, and divided into semantic chunks for efficient retrieval and accurate LLM responses.

6.3. Gantt Chart

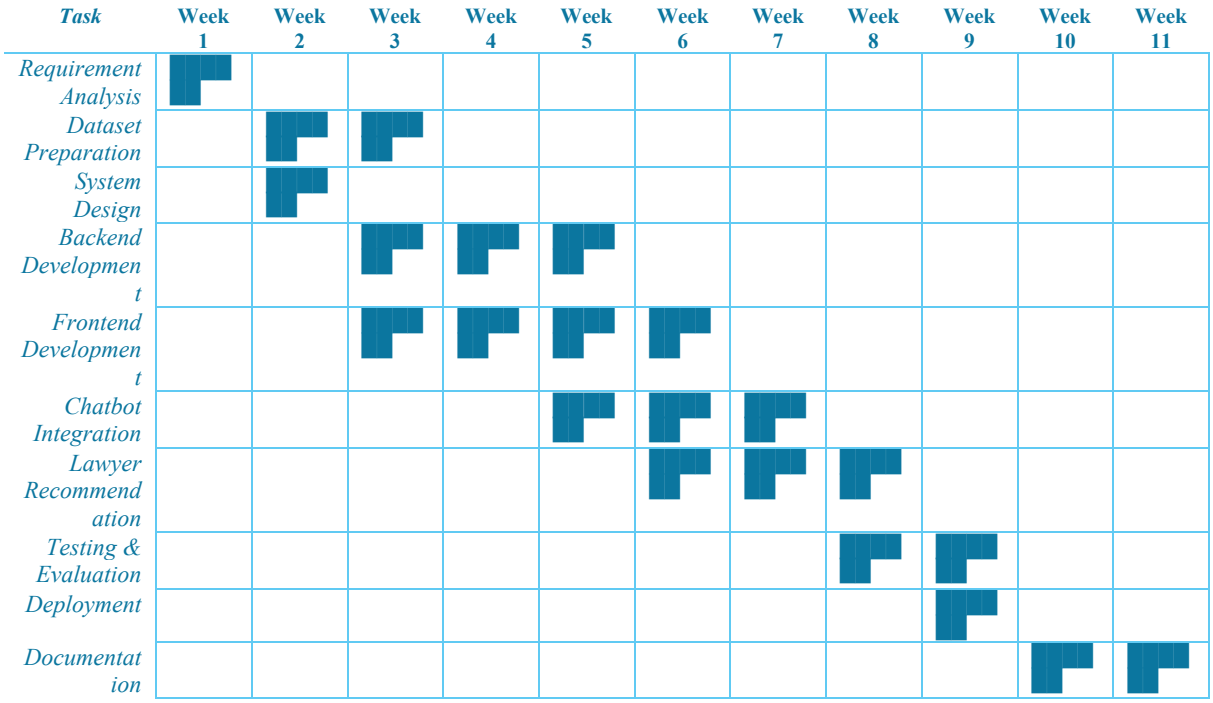

Figure 1.2

7. Project requirements

7.1. Functional requirements

Functional Requirement	Description
User Registration and Login	Applicants and employers can register and authenticate securely.
Resume Upload	Applicants can upload resumes (PDF/DOC/DOCX).
Resume Parsing	The system sends resumes to the AI model to extract skills, experience, and education.
Job Posting	Employers can add, edit, and view job vacancies.
Job Description Parsing	The AI model analyzes job requirements.
Matching & Scoring	The AI compares resumes and jobs to generate match scores.
Job Recommendations	Applicants can view a list of best-fit jobs ranked by match score.
Candidate Ranking	Employers can view ranked lists of applicants for each job.

7. 2. Non-functional requirements

Category	Requirement
Performance	Resume parsing and scoring should be completed within 5 seconds per request.
Scalability	Must support multiple concurrent uploads and job searches.
Security	Passwords encrypted; data transferred via HTTPS; role-based access.
Reliability	System uptime ≥ 99%; auto-recovery in case of crash.
Usability	Interface must be responsive and user-friendly on web and mobile browsers.

Maintainability	Codebase modularized into Web and AI components for independent updates.
Compatibility	Supports latest Chrome, Edge, and Firefox browsers.
Data Privacy	Personal information stored securely; user consent required for data sharing.

7.3. User requirements

User Type	Requirements
Applicant (Job Seeker)	1. Register and log in. 2. Upload or update resume. 3. View AI-based job recommendations. 4. Apply for selected jobs. 5. Receive notifications about application status.
Employer (Recruiter)	1. Register and log in. 2. Post and manage job listings. 3. View applicants ranked by match score. 4. Shortlist or rejected candidates. 5. Receive alerts when new relevant applicants apply.
System Administrator	1. Manage users and roles. 2. Monitor performance and logs. 3. Maintain AI model integration.

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Consumption;Machine Learning;Deep Learning;Feedforward Neural Networks;Random Forrest;Recurrent Neural Networks}